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Studies on 10kg alloy mass metal hydride based reactor for hydrogen storage

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HIGHLIGHTS

- A 10kg MH alloy reactor designed and fabricated.
- $LaNi_5$ alloy was used as the material for solid state hydrogen storage.
- Parametric Studies carried out with variation in key parameters.
- Optimum operational conditions for absorption & desorption determined.

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ABSTRACT

A 10 kg alloy mass metal hydride reactor, with $LaNi_5$ alloy was designed. Heat transfer enhacement in the reactor was achieved by including embedded cooling tubes and an external water jacket. Detailed parametric study has been carried to understand the performance of the system. The effect of both geometrical and operational parameters was studied in simulations. The optimized geometrical parameters were used for fabricating the reactor. Experimental studies were carried on the fabricated reactor. Absorption studies were carried out for different supply pressure and different cooling fluid temperatures. Storage capacity of 1.13 wt% was found in 1620 s at a supply pressure of 25 bar and with a flow rate of 20 LPM. Similarily, desorption studies were carried out for varying heat transfer fluid temperatures. Complete and fastest desorption was observed at 80 °C with the reaction completion time of 2700 s.

Abbreviations: BSP, British Standard Pipe; D, Dimensional; DAQ, Data Acquisition; ENG, Expanded Natural Graphite; ECT, Embedded Cooling Tubes; FS, Full Scale; GF, Gas Filter; HTF, Heat Transfer Fluid; LPM, Litres per Minute; MH, Metal Hydride; PCM, Phase Change Material; PR, Pressure Regulator; PT, Pressure Transducer; SS, Stainless Steel; TC, Thermocouple; V, Control Valve.

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Nomenclature

C_v	Heat Value (MJ/kg)
GC	Gravimetric Capacity (wt%)
L	Length (m)
M	Mass (kg)
P	Pressure (kPa)
T	Temperature (K)
t	Time (s)
Subscript	
a	Absorption
d	Desorption
m	Metal
s	Supply

Introduction

Today, the world is at a crossroads. It faces the uncertainty of balancing greenhouse gas emissions with the increasing global energy demands and at the same time the catastrophic climate change implications. The atmospheric CO_2 concentration in environment is increasing rapidly to significantly high levels due to extensive use of fossil fuels. The current challenge of decarbonizing the current industrial sector seems exceptionally complicated. The majority of renewable energy technologies are intermittent in nature, and thus replacing the use of oil and coal based power generation with such technologies seems challenging. In the quest for reliable green technologies for increasing renewable energy penetration, the large scale energy storage requirement is the primary bottleneck. Hydrogen can be considered as a potential energy carrier for storage and transmission of energy. Renewable energy technologies such as solar and wind can be integrated to utilize the surplus energy in hydrogen production through electrolysis, and hydrogen produced can be stored using different storage technologies for its efficient use later as and when desired.

Inspite of the high gravimetric energy density of hydrogen, there are numerous technological and economic barriers to large scale hydrogen technology deployment. Among the crucial challenges listed by US DOE are the development of efficient hydrogen storage technology, production of hydrogen using green technologies, delivery at low cost, and the development of cheaper and durable fuel cells for large scale applications. The first significant problem is the safe and efficient hydrogen storage due to its low hydrogen density among all gases. The two most common hydrogen storage methods deployed on a commercial scale are compressed hydrogen storage and liquefied hydrogen storage. In compressed hydrogen storage, hydrogen is stored at very high pressure in the range of 350–700 bar. Storage at such high pressures require specialized carbon fibre reinforced polymer based tanks, which significantly increases the storage cost besides this compression to such high pressures requires energy. Liquefied hydrogen storage offers better volumetric energy density than compressed storage, but liquefaction of hydrogen is a highly energy intensive process. Further storage of hydrogen in liquid state suffers from huge boil-off losses, which reduces the available hydrogen for the required applications. The third option available is in the form of solid state hydrogen storage which is very promising and safe method.

Metal hydrides offer high volumetric energy density. Hydrogen can be stored at lower pressures and at near ambient temperatures and thus this method has advantage as against compressed or liquid state storage. Apart from that, uptake and release can be easily controlled by controlling the operating parameters, making this method one of the most promising, reliable, practically scalable, and safer for being used for various applications. Because of all these positive features of metal hydrides, research for its commercial scalability has been carried out for several decades. Metal hydrides exhibit poor thermal conductivity, and charging and discharging of metal hydrides are exothermic and endothermic processes, respectively. Therefore, proper thermal management is crucial for reducing the cycle time of operation. Several studies on numerical and experimental research have been performed for improving the heat and mass transfer capability of metal hydride reactors with different heat exchanger designs.

In 1992, Ram Gopal et al. [1] developed a 1-D annular cylindrical model with an aim to create a generalized model for numerical analysis of different metal hydrides in dimensionless form. They formulated the correlations between space-averaged temperature and concentration in terms of metal hydride bed properties and operating parameters. These correlations were implemented using $LaNi_{4.7}Al_{0.3}$ bed and conclusions drawn were that the decreasing bed thickness and increasing thermal conductivity of metal hydride reactors results in improved performance. Ben and Jemni Nasrallah [2–4] developed a 2-D model for analyzing hydrogen absorption and desorption for a $LaNi_5$ based reactor. They validated the local thermal equilibrium hypothesis, negligibility of the effects of pressure variation in a reactor, and dependence of hydrogen concentration on equilibrium pressure. They also validated the simulation results experimentally using different operating parameters and found that improving the thermal conductivity of metal hydride bed results in optimal performance of storage device. In 2001, Mat and Kaplan [5], numerically and experimentally analyzed the hydride formation process in a metal hydride reactor with LmNi5 alloy. They reported that equilibrium pressure is an essential parameter for hydride formation. Reported results indicate that faster kinetics results in the low equilibrium pressure regions and an exponential increase in hydriding reaction in the beginning. The reaction was found to slow down as the bed temperature rises, due to exothermic reaction, which indicates that an effective heat transfer mechanism is required in a metal-hydride based hydrogen storage device to improve the reaction kinetics.

MacDonald and Andrew Rowe studied the effect of addition of external fins for thermal management using a 1-dimesional resistive and 2-dimensional transient numerical model for reactor with $LaNi_5$ alloy [6]. This study found that the addition of external fins has significant effect on heat transfer and the hydrogen pressure within the reactor. Mellouli et al. [7] developed a metal hydride tank with spiral heat exchanger and experimentally studied the absorption and desorption. Inclusion of heat exchanger design was reported to result in significant reduction of charge and discharge time. Melnihcuk et al. [8] numerically simulated and optimized the internal aluminum fin based heat exchanger design for MH tank filled with $LaNi_5$ alloy. This design was developed considering reactor diameter and absorption time as optimization parameters. A 2D numerical model was developed by Askri et al. [9], for different configurations of metal hydride reactors to study the dynamic behavior of metal hydride bed. The finding reported was that the reactor with concentric heat exchanger and fins provided with circulating HTF provision resulted in better charging time by 80% compared to other configurations. M. Raju et al. [10] developed Matlab Simulink Code based model in order to design a shell and tube heat exchanger based high-pressure metal hydride reactor. In the reactor developed by them, metal hydride was filled in shell region and heat transfer fluid flow was across the tubes. They analyzed refueling time and drive cycle for automotive applications and concluded that the reactor configuration with coolant at 360 LPM and 0 °C will be required in order to completely charge the tank within 5 min. Different configurations of metal hydride reactor design were studied by Visaria et al. [11] for reducing the charging time of the MH reactor. In order to achieve the target, they focused on the optimization of fins instead of cooling tubes. Kikkinides et al. [12] studied the effect of radius of cooling tubes and radial positions of a concentric annular ring within the reactor for efficient performance. The uniform arrangement of cooling tubes within reactor leads to optimal performance but then the selection of geometry of arrangement of these tubes plays a significant role. G. Mohan et al. [13] developed and optimized the metal hydride reactor with a different number of cooling tubes and porous filters for better heat transfer and reduced hydrogen charging time. However, in the various numerical studies carrieed some essential aspects such as convection term in the conservation of mass equation, pressure variation within bed, and hysteresis effect associated with metal hydrides were neglected. Mandhapati Raju et al. [14] performed 2D numerical studies using COMSOL Multiphysics and MATLAB software for the performance analysis and optimization of three different configurations of sodium alanate based hydrogen storage reactor. They concluded that a reactor with helical coil heat exchanger exhibits high heat transfer coefficient and the performance was better as compared to the shell and tube heat exchanger reactor design. Anbarasu et al. [15] performed an experimental study on two reactors with 36 and 60 ECT consisting of 2.75 kg of $LmNi_{4.91}Sn_{0.15}$ alloy to understand the effect of operating parameters on absorption rate and storage capacity. He observed that among the two configurations reactor with 60 ECT gets completely hydrogenated in 5 min at 30 °C, 25 bar supply pressure, and water flow rate of 30 LPM. Extensive numerical studies and experimental investigations have been done on using cooling tubes in the MH Reactors [10,17–19]. Literature reports suggest that embedded cooling tubes are far more efficient than air-cooled reactors having fins for the heat transfer.

Muthukumar et al. [20] developed a 2D mathematical model for performance analysis and optimization of large-scale metal hydride reactors with ECT and water jacket as heat exchanger, and misch-metal as hydrogen storage material. The geometrical optimization study for a cylindrical reactor with inner diameter of 50 mm, 20 ECT and a water jacket resulted into better absorption characteristics. Later as an extension to this work, they developed the optimized model for an industrial scale reactor to store 150 kg of $MmNi_{4.6}Al_{0.4}$ alloy. The reported finding was that a reactor with 300 mm diameter, 48 cooling tubes and a water jacket yields minimum absorption time at operating parameters of 30 bar, 20 °C, and 1250 W/m^2 K. In 2015, Bhouri et al. [21] developed a 2D model, where they analyzed and optimized the performance of a reactor with the combination of two hydrides i.e. metal hydride and complex hydride within a single reactor embedded with cooling tubes for enhanced heat transfer. The optimum configuration was a system with 49 hydride tubes and 95 cooling tubes, which was charged within 3 min. In 2018, Afzal et al. [22], presented a 2-dimensional numerical investigation of a metal hydride system capable of delivering 100 kWh energy using 210 kg Ti–Mn alloy. The

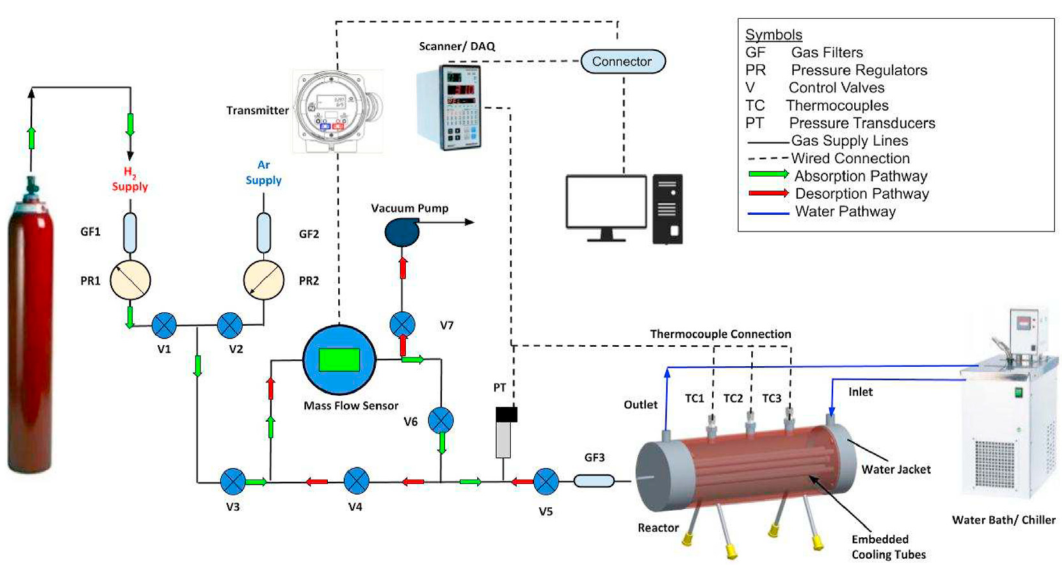

Fig. 1 – Schematic of the experimental set-up.

Fig. 2 – Left: Reactor assembly. Right: Gas manifold assembly.

optimum configuration was comprised of 14 multi-tubular MH bed in which the absorption reaction completes in 900 s and hydrogen gets discharged in 2000 s. Alok Kumar et al. [23], performed an experimental analysis of large scale metal hydride reactor developed for industrial applications with 99 embedded cooling tubes with 40 kg of $LaNi_{4.7}Al_{0.3}$ alloy with a gravimetric storage capacity of 1.4% capable of storing 552.4 g of hydrogen. They reported the effect of operational parameters such as pressure, temperature and heat transfer fluid flow rate. It took 1017s for 80% reaction fraction completion during absorption at 15 bar pressure, and heat transfer fluid maintained at 30 °C with flow rate being varied from 10 to 30 LPM, whereas desorption reaction took 3280 s for discharging 95% absorbed mass with bed temperature maintained at 80 °C. Broadly most of the reports available in literature have demonstrated different heat exchanger designs in order to improve heat transfer in the metal hydride beds. These techniques involve the use of cooling tubes, water jackets [24,25], a combination of cooling tubes and water jackets [12,26,27], phase change materials (PCM) [28–32], straight or longitudinal fins [11,27,33], pin fins, circular fins [9,24,34,35], combinations of fins and cooling tubes [7,9,11,13,15,16,22–25,33–36,43–50], staggered aluminum foils [8], zigzag aluminum foils [11], fins over spiral cooling tubes [36,37], the addition of foam and ENG [38–42].

Most of the research is focused on the design of lab-scale metal hydride based storage, while there are relatively very few reports available on commercial-scale metal hydride reactors. This gap needs to be bridged, considering the requirements for large scale adoption of hydrogen energy technologies and their widespread public acceptance. A significant amount of literature predicts that reactors with a combination of heat exchanger options such as embedded cooling tubes with fin or water jackets, lead to effective heat transfer solution for hydrogen storage. In our present work, with an intent to study the behavior of geometry and operational parameters on metal hydride bed in large-scale hydrogen storage reactors, we have simulated, designed, fabricated and performed experiments on a metal hydride reactor with 10 kg $LaNi_5$ alloy. The MH reactor model was analyzed with varying operational parameters for three different configurations i.e. (a) reactor without cooling tubes, (b) a reactor with cooling tubes and (c) a reactor with cooling tubes and an external water jacket, using COMSOL Multiphysics 5.3 software. Optimization of cooling tubes led to the conclusion that four cooling tubes with 0.0127 m inner diameter are sufficient for effective heat transfer and

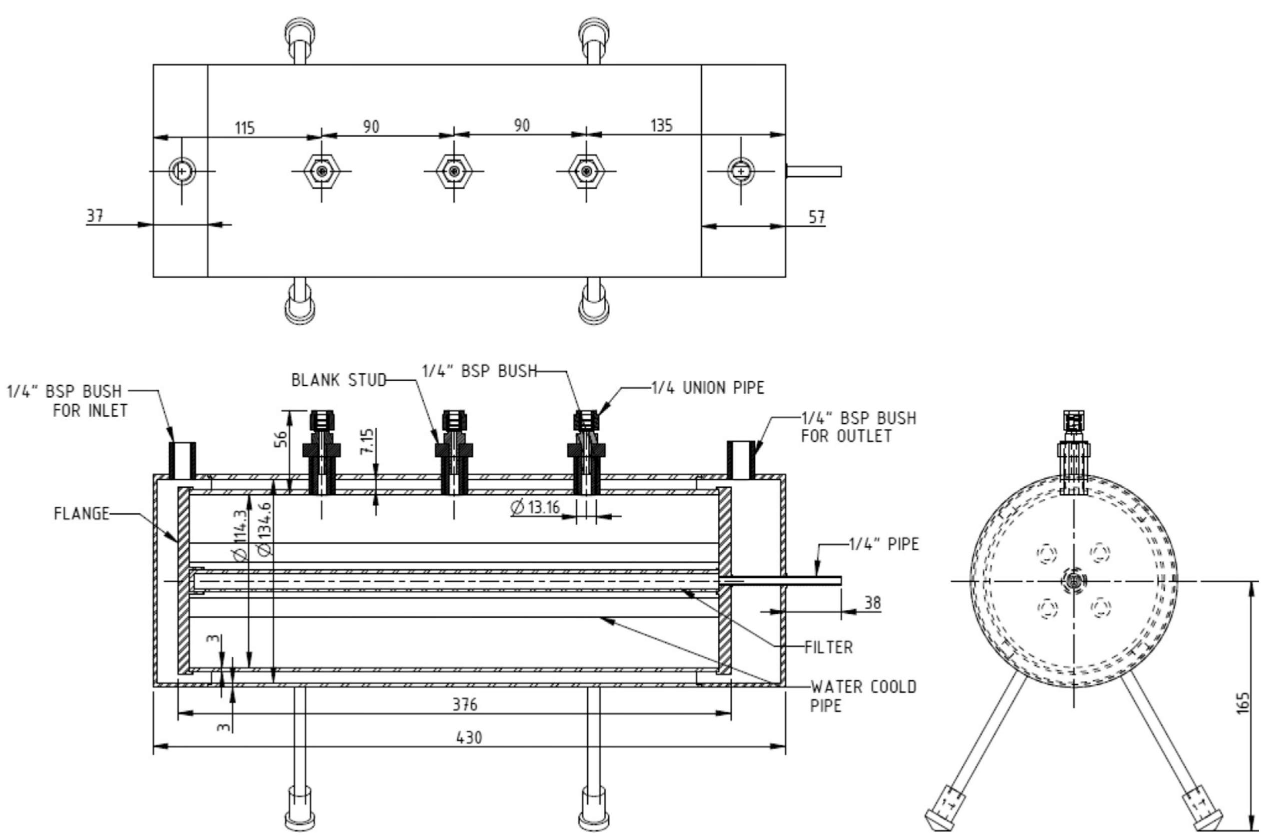

Fig. 3 – Dimensional specifications of reactor.

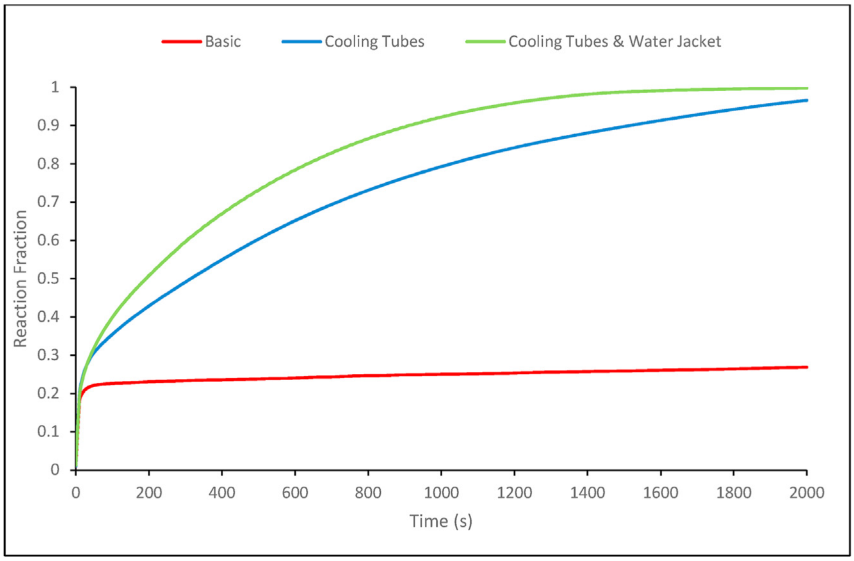

Fig. 4 – Comparison of absorption performance in different designs.

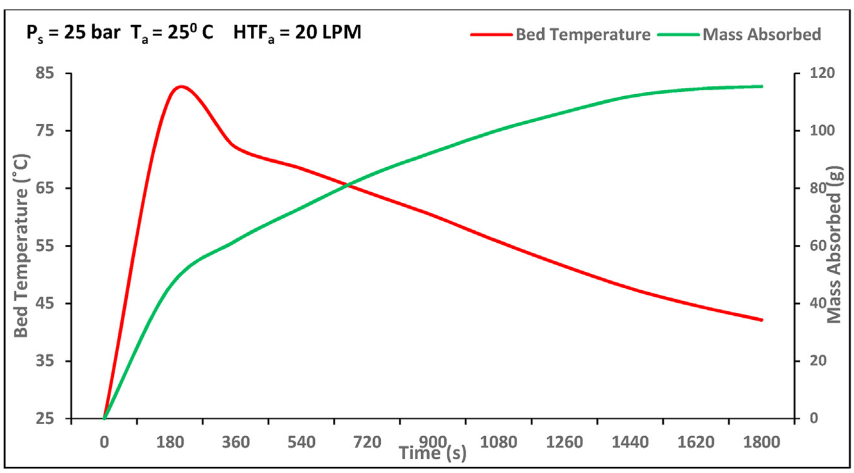

Fig. 5 – Variation of hydrogen absorbed and average bed temperature with time during the final activation cycle.

acceptable charging time. The simulation results suggested that a reactor with four cooling tubes and a water jacket yields much better performance compared to other configurations, which was then further taken for fabrication and experimental studies. The experimental investigations were performed with varying heat transfer fluid temperature and supply pressure to validate the numerical results. This work will be helpful in the development and scalability of metal hydride reactors.

Table 1 – Summary of parametric study conditions.

Half Cycle	Parameter	Value
Absorption	Supply Pressure (P_s)	10, 15, 20, 25 and 30 bar
	HTF Temperature (T_a)	5, 10, 20 and 30 °C
Desorption	HTF Temperature (T_d)	50, 60, 70 and 80 °C

Experimental set-up and reactor design

Description of experimental set-up

The system was designed such that gas lines are panel mounted. The system was divided into two sections, i.e., Gas manifold assembly and Reactor assembly. This arrangement enabled the use same setup for multiple experimentations, with only the reactors changed/replaced while the rest of the system can be same or remains unchanged. A schematic of the experimental setup is presented in Fig. 1.

The Gas manifold assembly and Reactor Assembly was integrated with a pressure transducer, thermocouples, Micro Motion mass flow meter, water bath/chiller, vacuum pump, a Data Acquisition Unit (DAQ), and a computer. A Baumer pressure transducer of range 0–100 bar & typical error ±0.2% of F.S, was used to measure the pressure. Three K-type thermocouples with a range of 0–200 °C were used to

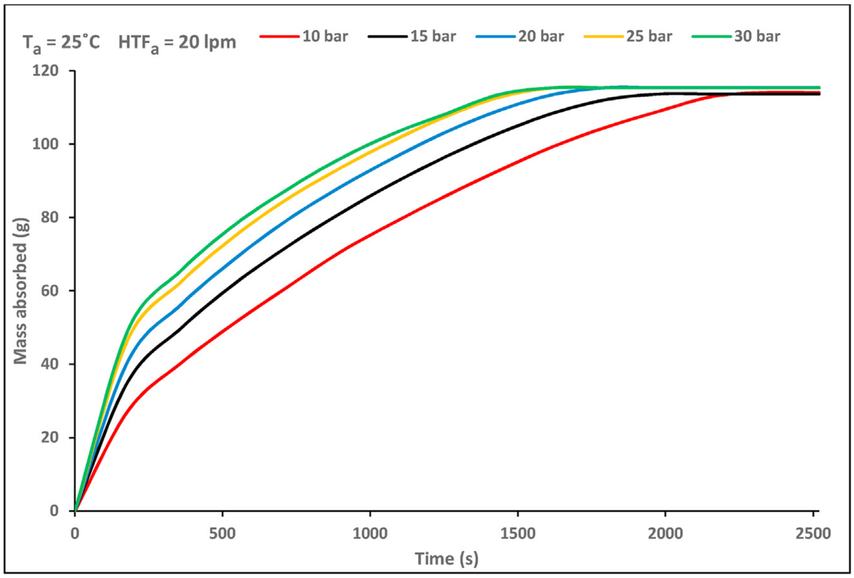

Fig. 6 – Effect on mass of hydrogen absorbed with varying supply pressure (P_s).

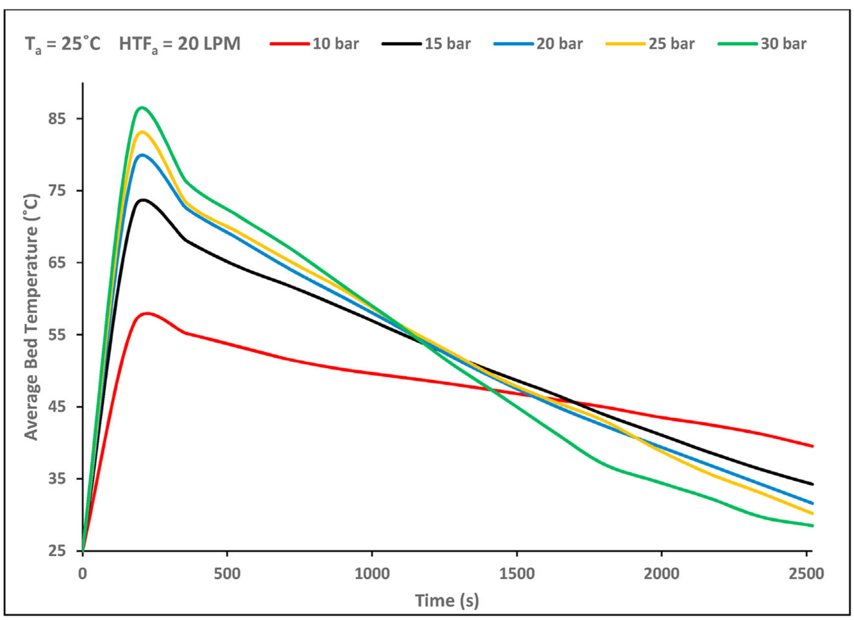

Fig. 7 – Effect of hydrogen supply pressure (P_s) on average bed temperature.

measure the temperature data at three different points along the reactor. The data from each was being recorded by the DAQ unit (Masibus 85XX+) using MSCAN + software. The mass flow rate of the hydrogen absorbed/desorbed was measured by a mass flow sensor connected to a transmitter (Micromotion Mass flow meter) with a mass flow accuracy of ±0.25% of the rate. Prolink III software was used to monitor the process and control flow parameters. The DAQ and the transmitter were integrated with a computer via Masibus connector (mUSB485). A 16 channel DAQ was used to record the data from the pressure transducer and thermocouples, and display onto the computer. The gas manifold included gas supply pipelines and a network of valves for inlet of hydrogen, argon, and evacuation through a vacuum pump as shown in Fig. 1. The reactor was supplied with water using a chiller (Escy CW-10HMP) with a maximum flow rate of 20 LPM and a temperature range of 0–80 °C. The water bath/chiller of accuracy ±0.1 °C and 5 L capacity and was used for the heat transfer fluid during the processes of absorption and desorption. A view of the gas manifold assembly and a picture of reactor is shown in Fig. 2. The 10 kg $LaNi_5$ system with gravimetric capacity of 1.2 wt% storing 0.12 kg of hydrogen was fabricated and thus gives a capacity of 4 kWh.

Details of the reactor design and specifications

The reactor was fabricated using SS316L to avoid embrittlement issues. The reactor has an outer diameter of 5.3 inches (0.134 m) and a length of 16.9 inches (0.430 m). A Schedule 40

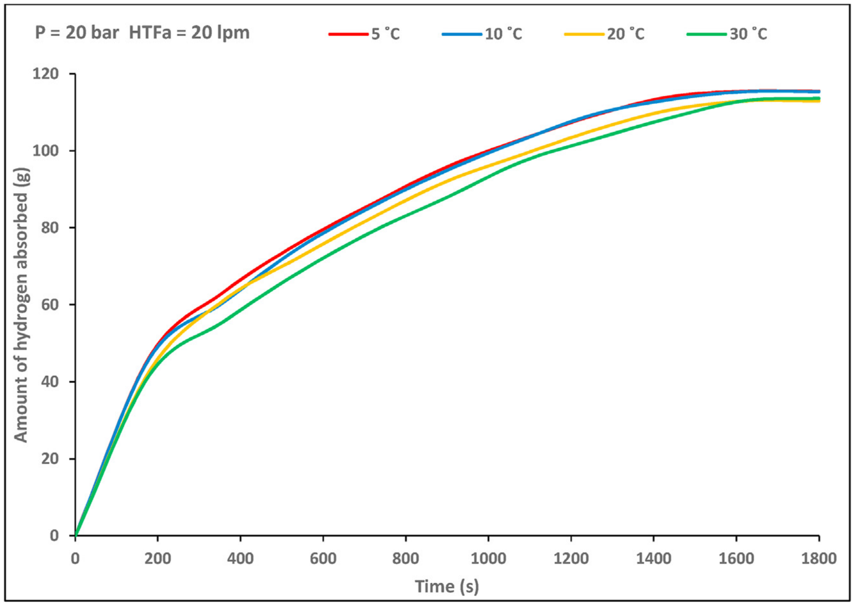

Fig. 8 — Amount of hydrogen absorbed for varying HTF temperature (T_a) during absorption.

pipe of 14.2 inches (0.36 m) length was selected. Three material filling points were provided using an adaptor opening of 0.5 inches (0.0127 m) each. The reactor primarily has two chambers — the outer chamber is where the cooling fluid flows, and the inner chamber housing the metal hydride alloy. The inner chamber is in its position through welded connections at eight locations with the outer chamber while maintaining sufficient cavity for water circualtion. There are five connections provided at the inlet and end flange of inner chamber — one in the center for hydrogen supply through a porous filter and four cooling tubes around periphery. A sintered SS metal filter was provided in the center of the reactor for hydrogen distribution across the length. The filter dimensions used in the system are - 0.01 m (inner diameter), 0.014 m (outer diameter), and 0.35 m length with pore size of 5 μm. The filter is blanked at one end and is welded to the reactor body on the other end. There are four cooling tubes around central porous filter. Three openings were provided for thermocouple probes and these were also used for material filling. Two water headers acting as inlet and outlet for heat transfer fluid were included. The reactor has four supports to enable the reactor be placed horizontally. The stand was provided with threaded connections at its end such that the reactor's height from the surface can be changed/altered whenever necessary. The dimensional specification of the reactor is shown in Fig. 3.

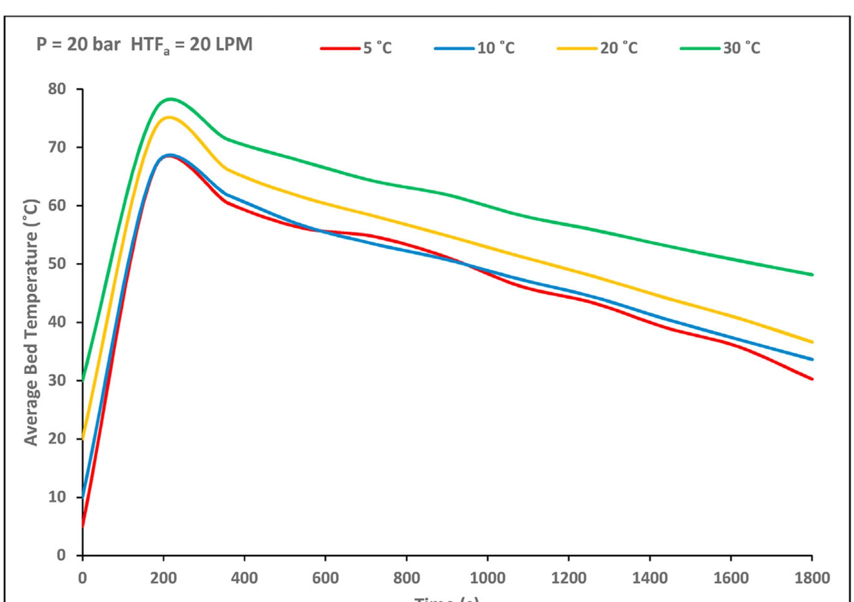

Fig. 9 — Effect of varying HTF temperature (T_a) on average bed temperature during.

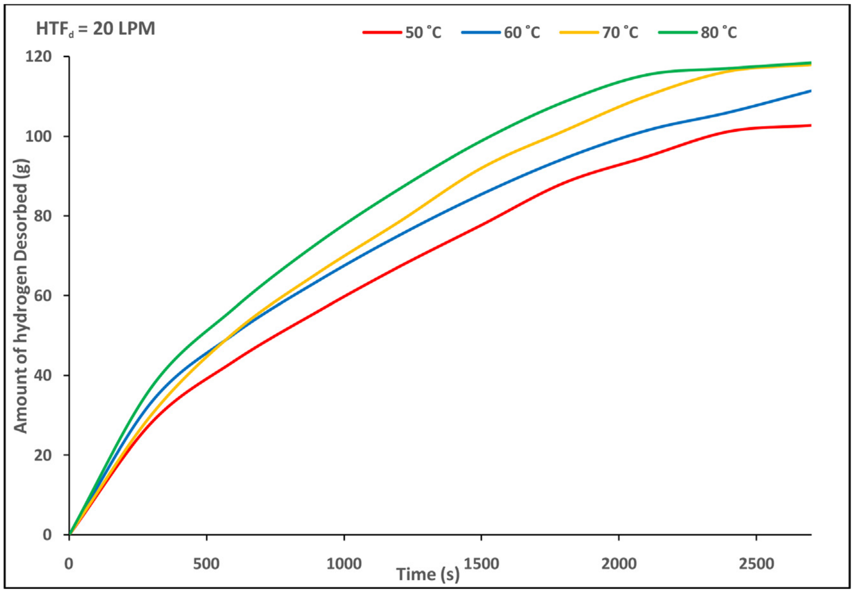

Fig. 10 – Amount of hydrogen desorbed with varying desorption temperature (T_d).

Simulation studies

Simulation studies on three different designs, i.e., a reactor without any heat exchanger (basic design), a reactor with four cooling tubes, and a reactor with four cooling tubes and a water jacket, was carried. As can be observed from Fig. 4, the reactor configuration with four cooling tubes and an external water jacket gives the best performance. The reaction completes in around 1650 s under 20 bar and a HTF flow rate of 20 LPM. The comparion of reaction fraction for three different designs is shown in Fig. 4.

Experimentation

Activation cycle

A thorough leak test was carried by pressurizing and leaving at that pressure, before material filling. The 10 kg $LaNi_5$ alloy

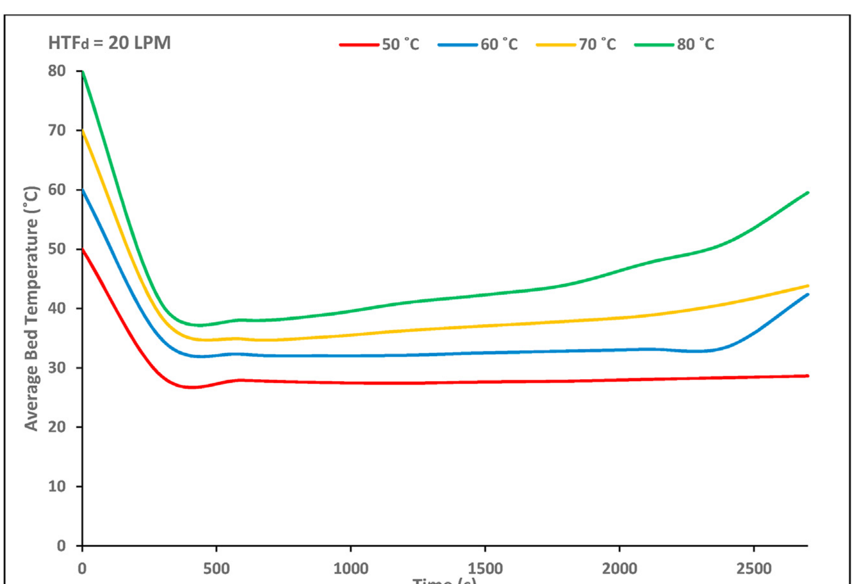

Fig. 11 – Variation of average bed temperature with varying desorption temperature (T_d).

was filled in the reactor using three filling ports. A series of steps was carried for the activation of the material. The reactor was first evacuated, then hydrogen was allowed into the system at 25 bar pressure. The material was then allowed to absorb hydrogen over 120 min. Any subsequent drop in pressure readings was noted. The entire system was heated up to 353 K and then desorbed which completed a cycle. Similar steps were repeated in different cycles. A capacity of 0.65, 1.01, and 1.03 was achieved in the third, fourth, and fifth cycles. This process was repeated for 15 cycles to ensure complete activation of the alloy. The final activation cycle showing the variation of mass of hydrogen absorbed and bed temperature at a supply pressure of 25 bar and HTF at 25 °C is presented in Fig. 5. A reversible capacity of 1.13 wt % was achieved.

Parametric Studies

The rate of absorption/desorption is dependent upon factors such as supply pressure and bed temperature. A parametric study includes studying the absorption and desorption cycles of the metal hydride bed under variations of these parameters. We aimed to ascertain the optimum conditions for the best performance of the system. The water flow rate into the reactor jacket inlet was at 20 LPM. Table 1 summarizes the experimental parameters for the study cycles. The results are being presented in the next section Results and discussion. The hydrogen flow rate cut-off was 0.027 g/s. The experimental uncertainty of the reactor's hydrogen storage capacity was determined using Kline & McClintock and estimated to be 3.5%.

Results and discussion

Absorption kinetics

Absorption is an exothermic process accompanied by the release of heat from the system. The absorption kinetics is affected by several factors such as supply pressure, heat transfer fluid temperature, convection coefficient, flow rate and others. The effect of variation of a few of the key major operating parameters has been studied and are being reported in this section. The average bed temperature is obtained by averaging the observations from the three thermocouples at different locations along the length of the reactor.

Effect of supply pressure (P_s)

The hydrogen supply pressure to the reactor was varied from 10 bar to 30 bar in steps of 5 bar. The pressure was maintained above the equilibrium pressure during absorption. The variation of pressure has a significant impact on the absorption time, as observed in Fig. 6. The operating conditions during the absorption process was a water flow rate of 20 LPM and temperature of HTF of 25 °C. The bed temperature rises instantly with the onset of the absorption process, as can been seen in Fig. 7. This rapid increase in temperature can be attributed to the metal hydride bed's low thermal conductivity and instantaneous heat accumulation in the beginning of the absorption reaction. The absorption process is governed by the driving force which is the difference between the hydrogen supply pressure and the metal hydride equilibrium pressure. A higher driving force leads to better and faster absorption. Absorption being an exothermic reaction, a sharp rise in temperature is observed in the beginning of the absorption process, as can be seen in Fig. 7. The high bed temperature leads to an increase in equilibrium pressure, and the driving force consequently decreases. This heat generated due to hydride forming reaction is removed by the external water jacket and embedded cooling tubes. Thus resulting in a continuous decrease in the bed temperature. The reaction slows down as the maximum reversible hydrogen storage capacity approaches, and we observe a plateau.

At supply conditions of 10 bar, 1.12 wt % hydrogen storage capacity was absorbed in 2340 s as observed in Fig. 6. The average bed temperature rises to 56.8 °C. As the supply pressure was increased to 15 bar, the reaction kinetics increased. Saturation of 1.12 wt % was achieved in 1980 s. The peak temperature achieved was 72.6 °C. Similarly, on increasing the supply pressure to 20 bar, the mass absorbed increased to 115.38 g in 1800 s. The peak temperature under this condition was observed to be 78.7 °C. The system stored 115.36 g of hydrogen in 1620 s at 25 bar and reached a plateau faster. The peak temperature was observed to be 81.8 °C at 25 bar. The progressive increase in pressure beyond 25 bar, not resulted in any significant change in the absorption time, as can be observed from Fig. 6. The reactor stored 115.37 g H_2 at 30 bar in 1620 s. The peak temperature increased to 85.16 °C at 30 bar. The performance of the system increased in terms of reduction in the absorption time with each 5 bar increase. At 25 bar, the reaction completion was achieved 30.76% faster than at 10 bar. However, the reaction completion time was similar for 25 and 30 bar with no significant increase in performance observed beyond 25 bar. Hence, we have considered 25 bar to be an optimum working pressure for the current reactor design.

Effect of variation of heat transfer fluid temperature (T_a) during absorption

The supply pressure of hydrogen was kept constant at 20 bar, while the heat transfer fluid temperature was varied to observe its effect on the bed temperature and amount of hydrogen absorbed. The absorption temperature was varied to 5, 10, 20, and 30 °C while the flow rate was kept constant at 20 LPM. The influence of absorption temperature on the mass of hydrogen absorbed and bed temperature is presented respectively in Figs. 8 and 9.

At 30 °C, a mass of 113 g was absorbed in 1800 s with a peak bed temperature of 76.4 °C. Although decreasing the absorption temperature to 20°C increased the rate of reaction initially, it took the same amount of time to attain a similar hydrogen storage capacity of 1.12 wt %. We observed a minor increase in the performance of the reactor when the temperature was at 10 °C. It absorbed 115.3 g of hydrogen in 1620 s and achieved a peak bed temperature of 40.5 °C. Meanwhile, the absorption temperature below 10 °C did not result in any further increase in performance.

Absorption process. A lower bed temperature favors lower equilibrium pressure and consequently aids the driving force at constant supply pressure. Hence, lower bed temperature increases the kinetics of the reaction. We observe that a lower heat transfer fluid temperature improves the performance time, but the impact is relatively less as against the cost it can add in reducing the temperature of HTF. Hence, absorption under room temperature is preferred.

Desorption kinetics

Desorption is the process of release of hydrogen from the metal hydride which was formed during the absorption, subjected to supplying heat. It is an endothermic phenomenon and energy is supplied to the reactor through the heat transfer fluid. An increase in bed temperature results in higher equilibrium pressure which leads to an increase in the differnce of equilibrium pressure and desorption pressure. This higher driving force and faster kinetics is favoured by increasing bed temperture. The HTF temperature was varied in steps of 10 °C from 50 °C to 80 °C. The flow rate was kept at a constant 20 LPM. The desorption studies i.e. the amount of mass desorbed and average bed temperature were studied under the mentioned conditions to determine the optimum operating conditions.

Effect of HTF temperature (T_d) during the desorption process
At an initial temperature of 50 °C, 102.75 g of hydrogen was desorbed in 2700 s while the bed temperature dropped to 28.5 °C. The bed temperature slowly rise to match the temperature of the transfer fluid in the later part of the reaction. An increase in the desorption temperature leads to faster release of hydrogen, as is evident from Fig. 10. Thus the time taken for complete desorption reduces as the desorption temperature increases. At 60 °C desorption temperature, the system desorbed 111.5 g of hydrogen in 2700s. The desorption being an endothermic process results in the drop in bed temperature, for a T_d of 60 °C, this drop was observed by 34.6 °C as seen in Fig. 11. With subsequent increase in desorption temperature to 70 °C, 118.05 g of hydrogen was desorbed in 2700 s and achieved a drop in bed temperature of 38.3 °C as shown in Fig. 11. There is an improvement observed in kinetics and the fastest desorption was observed to be achieved at 80 °C, resulting in complete desorption in 2700 s. Hence, desorption temperature has a significant impact on reducing desorption time and controlling the hydrogen release reaction.

Conclusion

A 10 kg alloy mass metal hydride reactor with $LaNi_5$ alloy was simulated, fabricated and experimentally studied. The material was activated, and a maximum reversible capacity of 1.13 wt % was achieved.

A parametric study was conducted to study the system's optimum operating conditions during both absorption and desorption. The major observations found during the study are listed below:

1. The supply pressure plays a significant role in determining the rate of absorption. At supply pressure of 10 bar, 1.12 wt % of hydrogen was achieved in 2340 s. The progressive increase in the pressure led to a decrease in reaction completion time compared to 10 bar, i.e., 1980 s at 15 bar, 1800 s at 20 bar, 1620 s at 25 bar. An increase in pressure above 25 bar lead to only slight improvement. The absorption kinetics studies shows that 25 bar is a favorable working pressure with a heat transfer fluid flow rate of 20 LPM.
2. The heat transfer fluid temperature was varied from 5 °C to 30 °C. A maximum hydrogen storage capacity of 1.11 wt % was achieved at both 30 °C and 20 °C in 1800 s. The storage capacity increased to 1.14 wt % at both 10 and 5 °C and the absorption time was found to be 1620 s. A lower temperature was found to favor a faster reaction, but the impact is not very significant, when compared in terms of the reaction completion time. Hence, ambient temperature, i.e., room temperature was preferred for carrying absorption than any lower temperature.
3. Desorption is an endothermic process and is aided by higher temperatures. The HTF temperature was varied between 50 °C and 80 °C. The desorption kinetics at 80 °C resulted in the fastest complete desorption, i.e., within 2700 s with a HTF flow rate at 20 LPM.

The experimental studies carried out in the current work are aimed at studying the effects of operational parameters in a large scale reactor. Its observed that the combined effect of embedded cooling tubes and water jacket is an efficient heat withdrawal measure and results in better performance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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